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Subobject Classifier [jc-0001]

November 3, 2026 · *Justin*

Outline [jc-0002]

1. Intro [jc-0003]

The subobject classifier seems to be very important for defining a topos, although I don't know what a topos is. In the process of understanding the definition of the subobject classifier, I think I gained more experience in playing around with representable functors, the Yoneda lemma, and natural transformations.

This post will introduce some concepts related to subobjects in a general category, and show how the subobject classifier ***internalizes*** these ideas. Many categorical definitions involving representable functors seem to be trying to internalize some external structure of the category; we will see more precise examples of this.

2. Generalized elements [jc-0004]

Let $\mathcal{C}$ be a category. In general, for an object A in $\mathcal{C}$, we can't talk about an element of A. But from the categorical semantics of type theory, or through the lens of the Yoneda lemma, it seems to be a great idea to consider the ***generalized elements*** of A.

Definition 2.1. generalized element [jc-0005]

Let $\mathcal{C}$ be a category and A an object of $\mathcal{C}$. A generalized element of A is a morphism into A

$$x : \Gamma \to A$$

where Γ, the source of x, is called the context of x. In the case where $\mathcal{C}$ has a terminal object 1 and the context of x is 1, x is called a global element of A.

The notation of generalized elements makes reasoning about objects very much like reasoning about "sets". The Yoneda embedding $\mathrm{y} : \mathcal{C} \to \mathrm{Psh}(\mathcal{C}), A \mapsto \hom_{\mathcal{C}}(-, A)$ can be viewed as converting an object into its generalized elements, sorted by context (where $\mathrm{Psh}(\mathcal{C}) := \mathrm{Set}^{\mathcal{C}^{\mathrm{op}}}$ is the presheaf category of $\mathcal{C}$, and we should assume $\mathcal{C}$ is locally small so that $\hom_{\mathcal{C}}(-, A)$ takes values in sets). The following exercise gives more intuition about the notation of generalized elements:

Exercise 2.2. [jc-0006]

Let $\mathcal{C}$ be a category and $f, g : X \to Y$ two morphisms in $\mathcal{C}$. Show that $f = g$ if and only if $fx = gx$ for every generalized element x of X (where fx denotes the composite $f \circ x$).

The name "global element" comes from the following understanding: if we have a global element $c : 1 \to A$, we can form the generalized element $c!_\Gamma : \Gamma \to A$ for each object Γ, where $!_\Gamma : \Gamma \to 1$ denotes the unique map into the terminal object. Such a collection of generalized elements $\{c!_\Gamma\}_{\Gamma \in \mathcal{C}}$ is invariant under "context substitution", in the sense that $c!_\Gamma \sigma = c!_\Delta$ for every morphism $\sigma : \Delta \to \Gamma$, since $!_\Gamma \sigma = !_\Delta$. Conversely, if we have a collection of generalized elements $\{c_\Gamma : \Gamma \to A\}_{\Gamma \in \mathcal{C}}$ invariant under context substitution, i.e. $c_\Gamma \sigma = c_\Delta$ for every morphism $\sigma : \Delta \to \Gamma$, then taking $c_1 : 1 \to A$, we have $c_1 !_\Gamma = c_\Gamma$ for each object Γ.

3. Subobjects [jc-0007]

To generalize subsets, we should look at monomorphisms $i : U \hookrightarrow A$. In the notation of generalized elements, the condition for $i : U \hookrightarrow A$ to be a monomorphism is: for all generalized elements x, y of U with the same context, if $ix = iy$ then $x = y$ (notice that ix and iy are generalized elements of A). So postcomposition with i can be seen as an inclusion of the generalized elements of U into those of A.

We can recover the notions of an element lying in a subset, of the inclusion relation between two subsets, and of when two subsets are considered equal.

Definition 3.1. [jc-0008]
For a monomorphism $u : U \hookrightarrow A$ and a generalized element $x : \Gamma \to A$ of A, we write $x \in u$ if x factors through u, i.e., there exists a morphism $\varphi : \Gamma \to U$ such that $x = u\varphi$.

Notice that such a φ is unique: if we have $\varphi, \psi : \Gamma \to U$ with $x = u\varphi$ and $x = u\psi$, then $u\varphi = u\psi$ implies $\varphi = \psi$ by the monicity of u. So it is appropriate that the notation $x \in u$ does not mention φ.

Definition 3.2. [jc-0009]
For monomorphisms $u : U \hookrightarrow A$ and $v : V \hookrightarrow A$, we write $u \leq v$ if for every generalized element x of A, $x \in u$ implies $x \in v$. We write $u \sim v$ if $u \leq v$ and $v \leq u$.

The following exercise shows that the definition of $u \leq v$ behaves as we expect and matches the definition on the nLab.

Exercise 3.3. [jc-000A]

1. $u \leq v$ iff u factors through v.
2. The relation $\leq$ on monomorphisms into A is a preorder.
3. The relation $\sim$ is an equivalence relation on monomorphisms into A.
4. $x \in u$ and $u \leq v$ are well defined with respect to the equivalence relation $\sim$; that is, if $u \sim u'$ then $x \in u$ iff $x \in u'$, and if $u \sim u'$ and $v \sim v'$ then $u \leq v$ iff $u' \leq v'$.

We close this section by giving the definition of the ***subobjects*** of an object A.

Definition 3.4. [jc-000B]
The subobjects of A are the equivalence classes of monomorphisms into A under $\sim$; in notation, $\mathrm{Sub}(A) := \{u \mid u : U \hookrightarrow A \text{ monic}\}/\sim$. If the collection $\mathrm{Sub}(A)$ is a set for every object A, the category is called well-powered.

4. The subobject functor and some examples [jc-000C]

We will look at some important examples of subobjects:

Example 4.1. [jc-000D]
Fix an object A. The identity $\mathrm{id}_A : A \hookrightarrow A$ is monic. It represents the maximum subobject of A, i.e. for every subobject u of A, we have $u \leq \mathrm{id}_A$. In other words, it "collects" all elements of A: for every generalized element x of A, we have $x \in \mathrm{id}_A$.

Example 4.2. [jc-000E]
Fix an object A. The diagonal embedding $\Delta_A : A \hookrightarrow A \times A$, $\Delta_A := \langle \mathrm{id}_A, \mathrm{id}_A \rangle$, is monic. It is an important step toward internalizing the equality of morphisms, in the sense that for all generalized elements x, y of A with the same context, we have $x = y$ if and only if $\langle x, y \rangle \in \Delta_A$.

Example 4.3. [jc-000F]
Let $f : A \to B$ be a morphism and $u : U \hookrightarrow B$ a subobject of B. A subobject $v : V \hookrightarrow A$ of A is called a preimage of u under f if for every generalized element x of A, $x \in v$ iff $fx \in u$.

One can check that the preimage of u under f is unique (up to the equivalence $\sim$ of subobjects) and well defined (it does not depend on the representative of the equivalence class of subobjects):

- If v, v' are preimages of u under f, then $v \sim v'$, since $x \in v$ iff $fx \in u$ iff $x \in v'$ for every generalized element x of A.
- If $u \sim u'$ and v, v' are preimages of u, u' under f respectively, then $v \sim v'$, since $x \in v \Leftrightarrow fx \in u \Leftrightarrow fx \in u' \Leftrightarrow x \in v'$ for every generalized element x of A.

So if a preimage of u under f exists, we can denote it by $f^{-1}(u) : f^{-1}(U) \hookrightarrow A$.

Taking preimages is functorial in the following sense (assuming all preimages exist):

- If u is a subobject of A, then $\mathrm{id}_A^{-1}(u) \sim u$, since $x \in \mathrm{id}_A^{-1}(u) \Leftrightarrow \mathrm{id}_A\, x \in u \Leftrightarrow x \in u$ for every generalized element x of A.
- If u is a subobject of C and $f : A \to B$, $g : B \to C$ are morphisms, then $(gf)^{-1}(u) \sim f^{-1}(g^{-1}(u))$, since $x \in (gf)^{-1}(u) \Leftrightarrow gfx \in u \Leftrightarrow fx \in g^{-1}(u) \Leftrightarrow x \in f^{-1}(g^{-1}(u))$ for every generalized element x of A.

How do we show that a preimage exists? For a generalized element $x : \Gamma \to A$, unpacking the condition $fx \in u$ gives $fx = uy$ for some $y : \Gamma \to U$, and we see a square:

$$\begin{array}{ccc} \Gamma & \xrightarrow{y} & U \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle u} \\ A & \xrightarrow[f]{} & B \end{array}$$

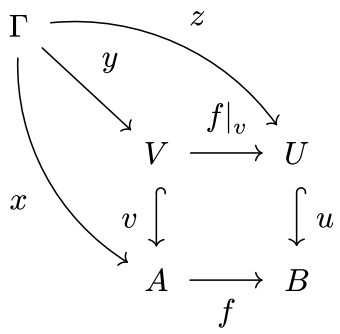
This gives me a strong belief that preimages are related to pullbacks. In the category of sets, the pullback of the inclusion map of a subset along a function is indeed the preimage. So we try:

If the category $\mathcal{C}$ has pullbacks, let $f : A \to B$ be a morphism and $u : U \hookrightarrow B$ a subobject of B. Consider the pullback square of u along f:

$$\begin{array}{ccc} V & \xrightarrow{f|_v} & U \\ {\scriptstyle v}\downarrow & & \downarrow{\scriptstyle u} \\ A & \xrightarrow[f]{} & B \end{array}$$

We want to show that $v : V \hookrightarrow A$ is the preimage of u under f. (Notice that the pullback of a monomorphism is again a monomorphism, so since u is monic, v is too.)

Let $x : \Gamma \to A$ be a generalized element of A. See the following diagram to keep track of the arrows:



- If $x \in v$, say $x = vy$ for some $y : \Gamma \to V$, then since the pullback square commutes, $fx = fvy = uf|_v\, y$, so $fx \in u$.
- Conversely, if $fx \in u$, say $fx = uz$ for some $z : \Gamma \to U$, then (Γ, x, z) forms a cone over the diagram $A \overset{f}{\to} B \overset{u}{\hookleftarrow} U$. By the universal property of the pullback V, there exists a unique $y : \Gamma \to V$ such that $x = vy$ and $z = f|_v\, y$. So $x \in v$. ∎

Exercise 4.4. [jc-000G]
Conversely, without assuming that the category has pullbacks: if $v : V \hookrightarrow A$ is the preimage of u under f, then $v = v\,\mathrm{id}_V$, so $v \in v$; since v is the preimage, $fv \in u$, and we get $fv = ug$ for some $g : V \to U$. Show that (V, v, g) is the pullback of the diagram $A \overset{f}{\to} B \overset{u}{\hookleftarrow} U$.

We summarize this example with the following proposition:

Proposition 4.5. Subobject functor [jc-000H]
Let $\mathcal{C}$ be a well-powered category with pullbacks. We have the subobject functor

$$\begin{aligned} \mathrm{Sub} : \mathcal{C}^{\mathrm{op}} &\to \mathrm{Set} \\ A &\mapsto \mathrm{Sub}(A) \\ f : A \to B &\mapsto f^{-1} : \mathrm{Sub}(B) \to \mathrm{Sub}(A) \end{aligned}$$

mapping an object A to the set of all subobjects of A, and a morphism $f : A \to B$ to the preimage function $f^{-1} : \mathrm{Sub}(B) \to \mathrm{Sub}(A)$, which sends a subobject $u \in \mathrm{Sub}(B)$ of B to the unique subobject $f^{-1}(u) \in \mathrm{Sub}(A)$ of A such that $x \in f^{-1}(u) \Leftrightarrow fx \in u$ for every generalized element x of A. Equivalently, $f^{-1}(u)$ is defined as the pullback of u along f:

$$\begin{array}{ccc} f^{-1}(U) & \longrightarrow & U \\ {\scriptstyle f^{-1}(u)}\downarrow & & \downarrow{\scriptstyle u} \\ A & \xrightarrow[f]{} & B \end{array}$$

Key Exercise 4.6. [jc-000I]

Assume pullbacks exist, and let u be a subobject of A. Show that $x \in u$ iff $x^{-1}(u) \sim \mathrm{id}_\Gamma$ for every generalized element $x : \Gamma \to A$ of A. This says that every proposition like $x \in u$ can be written as an equality of subobjects. Using the diagonal embedding above ($x = y$ iff $\langle x, y \rangle \in \Delta_A$), the primitive predicates of the language become equalities of subobjects!

Example 4.7. [jc-000J]
Assume the category has pullbacks. Let $u : U \hookrightarrow A$ and $v : V \hookrightarrow A$ be subobjects of A. A subobject w of A is an intersection of u and v if $x \in w$ iff $x \in u$ and $x \in v$. An intersection, if it exists, is unique and well defined. Consider the pullback of v along u:

$$\begin{array}{ccc} & & u|_v \\ & u^{-1}(V) & \hookrightarrow & V \\ u^{-1}(v) & \downarrow & & \downarrow v \\ & U & \underset{u}{\hookrightarrow} & A \end{array}$$

Then the subobject of A given by the composite $u^{-1}(V) \overset{u^{-1}(v)}{\hookrightarrow} U \overset{u}{\hookrightarrow} A$ is the intersection of u and v, denoted by $u \cap v$.

5. The greatest subobject as a natural transformation $\{*\} \to \mathrm{Sub}$ [jc-000K]

- The pullback of the identity.
- Naturality: essentially, there is only one construction.
- In the presence of a terminal object, a uniform choice of subobjects corresponds to a subobject of 1.
- The Yoneda lemma: find the one essential construction, the universal element $\mathrm{id}_1 : 1 \hookrightarrow 1$.

Now that we have established the subobject functor, before we step into the subobject classifier, I want to change perspective and look at the greatest subobject $\mathrm{id}_A : A \hookrightarrow A$ again.

The construction of $\mathrm{id}_A : A \hookrightarrow A$ does not seem to depend on the object A: we can do this "uniform" construction for every object. And maybe there is only one "essential" subobject.

This construction is "uniform" in the sense that we have $\mathrm{id}_A \in \mathrm{Sub}(A)$ for each object A, and the functoriality of Sub preserves id_A; namely, for any $f : A \to B$, we have $f^{-1}(\mathrm{id}_B) \sim \mathrm{id}_A$.

$$\begin{array}{ccccc} B & & \mathrm{Sub}(B) & & \mathrm{id}_B \\ f \uparrow & & f^{-1} \downarrow & & \downarrow \\ A & & \mathrm{Sub}(A) & & \mathrm{id}_A \sim f^{-1}(\mathrm{id}_B) \end{array}$$

Indeed, for every generalized element x of A, we have $x = \mathrm{id}_A\, x$ and $fx = \mathrm{id}_B\, fx$, so both $x \in \mathrm{id}_A$ and $fx \in \mathrm{id}_B$ (i.e. $x \in f^{-1}(\mathrm{id}_B)$) hold; hence $\mathrm{id}_A \sim f^{-1}(\mathrm{id}_B)$.

This is exactly the idea of a natural transformation! Let $\underline{\{*\}} : \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ be the contravariant constant functor sending every object to the singleton set $\{*\}$ and every morphism to the identity $\mathrm{id}_{\{*\}}$ on the singleton set. We can state the uniformity of picking the subobject id_A for each object A as follows: we have a natural transformation defined by

$$\begin{aligned} \eta &: \underline{\{*\}} \to \mathrm{Sub} \\ \eta_A &: \{*\} \to \mathrm{Sub}(A) \\ \eta_A &: * \mapsto \mathrm{id}_A \end{aligned}$$

Naturality says that for any $f : A \to B$, the following diagram commutes:

$$\begin{array}{ccccc} B & & \{*\} & \xrightarrow{\eta_B} & \mathrm{Sub}(B) \\ f \uparrow & & \mathrm{id}_{\{*\}} \| & & f^{-1} \downarrow \\ A & & \{*\} & \xrightarrow[\eta_A]{} & \mathrm{Sub}(A) \end{array} \qquad \begin{array}{ccc} * & \longmapsto & \mathrm{id}_B \\ \downarrow & & \downarrow \\ * & \longmapsto & \mathrm{id}_A \sim f^{-1}(\mathrm{id}_B) \end{array}$$

so $\mathrm{id}_A \sim f^{-1}(\mathrm{id}_B)$!

Naturality describes the "uniform" part, and the Yoneda lemma helps us find the "essential" core. Recall the definition of the terminal object 1: for any object A, there is one and only one arrow $!_A : A \to 1$ from A into 1; in other words, $\mathrm{hom}_{\mathcal{C}}(A, 1)$ is a singleton set, so we have a bijection of sets

$$\{*\} \cong \mathrm{hom}_{\mathcal{C}}(A, 1), * \mapsto !_A$$

Moreover, this bijection is natural in A (the assignment of $!_A$ to each object A is also uniform); that is, the following defines a natural transformation (in fact, a natural isomorphism, since every component is a bijection):

$$\begin{aligned} \xi &: \underline{\{*\}} \cong \mathrm{hom}_{\mathcal{C}}(-, 1) \\ \xi_A &: \{*\} \cong \mathrm{hom}_{\mathcal{C}}(A, 1) \\ \xi_A &: * \mapsto !_A \end{aligned}$$

Using the fact that the functoriality of $\mathrm{hom}_{\mathcal{C}}(-, 1)$ is given by precomposition $(\circ f) : \mathrm{hom}_{\mathcal{C}}(B, 1) \to \mathrm{hom}_{\mathcal{C}}(A, 1)$, one can check that the naturality of ξ says $!_A = !_B f$ for every $f : A \to B$. Composing them, we get a natural transformation

$$\theta := \eta\xi^{-1} : \hom_{\mathcal{C}}(-,1) \overset{\xi^{-1}}{\cong} \underline{\{*\}} \overset{\eta}{\to} \mathrm{Sub}$$

whose component at A is $\hom_{\mathcal{C}}(A,1) \to \mathrm{Sub}(A), !_A \mapsto \mathrm{id}_A$.

Think about how to make a uniform mapping that sends a morphism $f : A \to 1$ to a subobject $u_f \in \mathrm{Sub}(A)$ of A, where the mapping has to work for every object A. If we have a subobject $v \in \mathrm{Sub}(1)$ of 1, we can use the functoriality of Sub to get $f^{-1}(v) \in \mathrm{Sub}(A)$. It turns out that this is the only way to make such a uniform mapping! The trick is that we can use $\mathrm{id}_1 : 1 \to 1$ to detect such a uniform mapping!

Consider the component of $\theta : \hom_{\mathcal{C}}(-,1) \to \mathrm{Sub}$ at 1, with input id_1:

$$\theta_1 : \hom_{\mathcal{C}}(1,1) \to \mathrm{Sub}(1)$$
$$\theta_1 : \mathrm{id}_1 = !_1 \mapsto \mathrm{id}_1$$

(On the left-hand side, $\mathrm{id}_1 : 1 \to 1$ is a morphism; on the right-hand side, $\mathrm{id}_1 : 1 \hookrightarrow 1$ is a subobject in $\mathrm{Sub}(1)$.)
Naturality recovers every component from the "essential" subobject $\mathrm{id}_1 \in \mathrm{Sub}(1)$: for any object A, we have the morphism $!_A : A \to 1$, and by naturality the following diagram commutes:

$$\begin{array}{ccc} 1 & \hom_{\mathcal{C}}(1,1) \xrightarrow{\theta_1} \mathrm{Sub}(1) & \mathrm{id}_1 = !_1 \longmapsto \mathrm{id}_1 \\ {\scriptstyle !_A}\uparrow & \downarrow{\scriptstyle (\circ\, !_A)} \quad {\scriptstyle !_A^{-1}}\downarrow & \downarrow \quad\quad \downarrow \\ A & \hom_{\mathcal{C}}(A,1) \xrightarrow[\theta_A]{} \mathrm{Sub}(A) & \mathrm{id}_1\, !_A = !_A \longmapsto \mathrm{id}_A \sim !_A^{-1}(\mathrm{id}_1) \end{array}$$

so $\mathrm{id}_A \sim !_A^{-1}(\mathrm{id}_1) \in \mathrm{Sub}(A)$. Every greatest subobject $\mathrm{id}_A \in \mathrm{Sub}(A)$ is a *shadow* of the "essential" core $\mathrm{id}_1 \in \mathrm{Sub}(1)$, in the sense that id_A is the preimage of id_1 under $!_A$ for every object A.

This process of finding the "essential" core is basically the proof of the Yoneda lemma and the idea behind representable functors. We state three theorems and definitions that we will use in the following paragraphs. I suggest also looking at the nLab for more detail, but don't worry about these statements: we will unpack the proofs wherever we use these theorems.

Theorem 5.1. Yoneda lemma [jc-000L]
Let $\mathcal{C}$ be a locally small category, $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ a contravariant set-valued functor, and X an object of $\mathcal{C}$. Then we have a bijection

$$\hom_{\mathrm{Psh}(\mathcal{C})}(\hom_{\mathcal{C}}(-,X), F) \cong F(X)$$

which sends a natural transformation $\eta \in \hom_{\mathrm{Psh}(\mathcal{C})}(\hom_{\mathcal{C}}(-, X), F)$, i.e. $\eta : \hom_{\mathcal{C}}(-, X) \to F$, to the element $\eta_X(\mathrm{id}_X) \in F(X)$ (the essential core!). Here $\mathrm{Psh}(\mathcal{C}) := \mathrm{Set}^{\mathcal{C}^{\mathrm{op}}}$ denotes the presheaf category, whose objects are functors and whose morphisms are natural transformations. There is also a version for covariant set-valued functors $G : \mathcal{C} \to \mathrm{Set}$.

Definition 5.2. Representable functor [jc-000M]
Let $\mathcal{C}$ be a locally small category. A functor $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ is called representable if there exist an object X in $\mathcal{C}$ and a natural isomorphism

$$\theta : \hom_{\mathcal{C}}(-, X) \cong F$$

In this case, X is called the representing object of F. By the Yoneda lemma, θ is uniquely determined by an element $e \in F(X)$, called the universal element of F. There is also a version for covariant set-valued functors $G : \mathcal{C} \to \mathrm{Set}$.

Corollary 5.3. Yoneda embedding [jc-000N]
Let $\mathcal{C}$ be a locally small category and X, Y two objects of $\mathcal{C}$. If we have an isomorphism

$$\hom_{\mathcal{C}}(\Gamma, X) \cong \hom_{\mathcal{C}}(\Gamma, Y)$$

natural in Γ (i.e. a natural isomorphism $\hom_{\mathcal{C}}(-, X) \cong \hom_{\mathcal{C}}(-, Y)$), then $X \cong Y$. In the notation of generalized elements, it informally says that if X and Y have the same generalized elements, then X and Y are the same.

Exercise 5.4. [jc-0000]
In the opening section, we explained why a morphism $1 \to A$ out of the terminal object is called a global element. Revisit that section, rewrite the condition "invariant under context substitution" as a natural transformation, and show that we have the bijection

$$\hom_{\mathrm{Psh}(\mathcal{C})}(\underline{\{*\}}, \hom_{\mathcal{C}}(-, A)) \cong \hom_{\mathcal{C}}(1, A)$$

(Bonus: it is also natural in A.)

6. Defining the subobject classifier [jc-000P]

- Representable functors: internalization.
- Indicator functions; unpacking naturality.
- The Yoneda lemma: find the one essential construction.
- Compute the inclusion map of the universal subobject $T \hookrightarrow \Omega$.

Now we have enough courage to look at the definition of the subobject classifier. From now on, we

assume that our category $\mathcal{C}$ has pullbacks and a terminal object, and is locally small and well-powered.

Definition 6.1. [jc-000Q]
The subobject classifier Ω is the representing object of the subobject functor Sub. Explicitly, Ω is an object of $\mathcal{C}$ together with a natural isomorphism

$$\chi : \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega)$$

From the discussion above, we expect the naturality of χ to give us a uniform way to convert a subobject of A into a morphism into Ω. We may also use the Yoneda lemma to find the essential core of χ. Moreover, the isomorphism part of χ says that every subobject is completely characterized by an arrow into Ω. We encourage the reader to play around with the naturality of χ and see what the examples in the previous sections correspond to; you may be able to guess all the constructions we will do in the sections below.

Let's unpack the definition of the subobject classifier:

$$\begin{aligned} \chi : &\quad \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega) \\ \chi_A : &\mathrm{Sub}(A) \cong \hom_{\mathcal{C}}(A, \Omega) \\ \chi_A : &\qquad u \mapsto \chi_A(u) =: \chi_u \end{aligned}$$

First, χ is a natural transformation; it has a component $\chi_A : \mathrm{Sub}(A) \to \hom_{\mathcal{C}}(A, \Omega)$ for each object A. Let's denote the image of a subobject $u \in \mathrm{Sub}(A)$ under χ_A by $\chi_u := \chi_A(u)$. Notice that we can infer the object A from the target of the monomorphism $u : U \hookrightarrow A$, so it is appropriate that the notation χ_u does not mention A. We call χ_u the indicator (or characteristic) function of u.

The naturality of χ gives (can you guess what it is just by recalling the functoriality of Sub and $\hom_{\mathcal{C}}(-, \Omega)$, without drawing the naturality square?), for any morphism $f : A \to B$, that

$$\begin{array}{ccc} B & \mathrm{Sub}(B) & \xrightarrow{\chi_B} \hom_{\mathcal{C}}(B, \Omega) \\ f \uparrow & f^{-1} \downarrow & (\circ f) \downarrow \\ A & \mathrm{Sub}(A) & \xrightarrow[\chi_A]{} \hom_{\mathcal{C}}(A, \Omega) \end{array} \qquad \begin{array}{ccc} u & \longmapsto & \chi_u \\ \downarrow & & \downarrow \\ f^{-1}(u) & \longmapsto & \chi_{f^{-1}(u)} = \chi_u f \end{array}$$

commutes, that is, $\chi_{f^{-1}(u)} = \chi_u f$ for every subobject $u \in \mathrm{Sub}(B)$. (Compare this with the set-theoretic indicator function of a subset!)

Since χ is an isomorphism, every morphism into Ω should be the indicator function of some subobject. In fact, we can compute this subobject explicitly. The Yoneda lemma tells us to consider the universal subobject $t : T \hookrightarrow \Omega \in \mathrm{Sub}(\Omega)$, defined by

$$\chi_\Omega : \mathrm{Sub}(\Omega) \cong \hom_{\mathcal{C}}(\Omega, \Omega)$$
$$\chi_\Omega : t \mapsto \mathrm{id}_\Omega$$

For any morphism $f : A \to \Omega$, applying naturality to $f : A \to \Omega$ and $t \in \mathrm{Sub}(\Omega)$ gives

$$\begin{aligned} \chi_{f^{-1}(t)} &= \chi_t f \\ &= \mathrm{id}_\Omega \, f \\ &= f \end{aligned}$$

That is, $\hom_{\mathcal{C}}(A, \Omega) \to \mathrm{Sub}(A), f \mapsto f^{-1}(t)$ is the inverse of χ: every subobject is just a preimage of the universal subobject t!

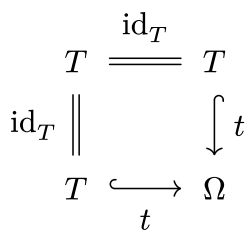
Key Exercise 6.2. [jc-000R]
Show that $x \in u \Leftrightarrow \chi_u x \in t$ for every subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A. This justifies calling χ_u an indicator function, and naming t after the first letter of "true".

Finally, we can also compute the monomorphism $t : T \hookrightarrow \Omega$ as the indicator function of some subobject. To show off the proof techniques of category theory, we compute $t^{-1}(t)$ in three ways:

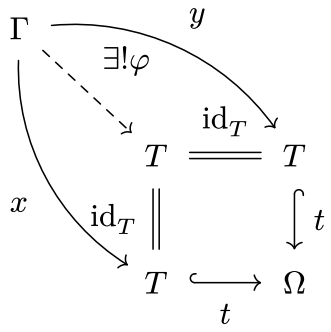
- Computing $t^{-1}(t)$ by pullback: consider a cone $T \xleftarrow{x} \Gamma \xrightarrow{y} T$ over the diagram $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$:

$$\begin{array}{ccc} \Gamma & \xrightarrow{y} & T \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle t} \\ T & \underset{t}{\hookrightarrow} & \Omega \end{array}$$

  We have two pieces of data, x and y, and one constraint, $tx = ty$. Since t is monic, $x = y$. So a cone over $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$ really consists of one piece of data, $x : \Gamma \to T$, without any constraint; it holds the same data as a cone over the diagram consisting of the single object T. But the limit of the diagram consisting of the single object T is T with the universal cone $\mathrm{id}_T : T \to T$. So we claim that the following is a pullback square:



Indeed, it commutes: $t\,\mathrm{id}_T = t\,\mathrm{id}_T$. Given any cone $T \xleftarrow{x} \Gamma \xrightarrow{y} T$ with $tx = ty$:



since $tx = ty$ and t is monic, $x = y$.

- Existence of φ: take $\varphi := x : \Gamma \to T$; then $x = \mathrm{id}_T\, x$ and $y = x = \mathrm{id}_T\, x$.
- Uniqueness of φ: if $\psi : \Gamma \to T$ satisfies $x = \mathrm{id}_T\, \psi$ and $y = \mathrm{id}_T\, \psi$, then $x = \mathrm{id}_T\, \psi = \psi$. (In general, the pullback of a monomorphism along itself is the identity map.)

Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

- Computing $t^{-1}(t)$ by the Yoneda lemma: in general, the existence of a pullback of the diagram $A \xrightarrow{f} C \xleftarrow{g} B$ says exactly that we have an object $A \times_C B$ and an isomorphism

$$\hom_{\mathcal{C}}(\Gamma, A \times_C B) \cong \{(x, y) \in \hom_{\mathcal{C}}(\Gamma, A) \times \hom_{\mathcal{C}}(\Gamma, B) \mid fx = gy\}$$

  natural in Γ. Don't be scared by this statement. Think about the functoriality of the functors on both sides (verify that the right-hand side is a functor in Γ! It sends an object Γ to the collection of cones over the diagram $A \xrightarrow{f} C \xleftarrow{g} B$ with apex Γ, and it acts on morphisms by precomposition). Follow the process above to unpack the naturality and find the universal element; you will see that it is exactly the universal property of the pullback.

To compute the pullback $T \times_\Omega T$ of the diagram $T \overset{t}{\hookrightarrow} \Omega \overset{t}{\hookleftarrow} T$, the machinery of the Yoneda lemma and representable functors lets us do the following computation:

$$
\begin{aligned}
&\hom_{\mathcal{C}}(\Gamma, T \times_\Omega T) \\
\cong&\{(x, y) \in \hom_{\mathcal{C}}(\Gamma, T) \times \hom_{\mathcal{C}}(\Gamma, T) \mid tx = ty\} \\
\cong&\{(x, y) \in \hom_{\mathcal{C}}(\Gamma, T) \times \hom_{\mathcal{C}}(\Gamma, T) \mid x = y\} \\
\cong&\{x \in \hom_{\mathcal{C}}(\Gamma, T) \mid \mathrm{True}\} \\
=&\hom_{\mathcal{C}}(\Gamma, T)
\end{aligned}
$$

This is an isomorphism natural in Γ (why?). So by the Yoneda lemma, $T \times_\Omega T \cong T$, and the universal cone is given by the element corresponding to $\mathrm{id}_T \in \hom_{\mathcal{C}}(T, T)$ (why?):

$$
\begin{aligned}
&\{(x, y) \in \hom_{\mathcal{C}}(T, T) \times \hom_{\mathcal{C}}(T, T) \mid tx = ty\} &&\ni (\mathrm{id}_T, \mathrm{id}_T) \\
\cong&\{(x, y) \in \hom_{\mathcal{C}}(T, T) \times \hom_{\mathcal{C}}(T, T) \mid x = y\} &&\ni (\mathrm{id}_T, \mathrm{id}_T) \\
\cong&\{x \in \hom_{\mathcal{C}}(T, T) \mid \mathrm{True}\} &&\ni \mathrm{id}_T \\
=&\hom_{\mathcal{C}}(T, T) &&\ni \mathrm{id}_T
\end{aligned}
$$

so the following is a pullback square:

$$
\begin{array}{ccc}
T & \overset{\mathrm{id}_T}{=\!=} & T \\
{\scriptstyle \mathrm{id}_T}\,\| & & \downarrow{\scriptstyle t} \\
T & \underset{t}{\hookrightarrow} & \Omega
\end{array}
$$

Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

- Computing $t^{-1}(t)$ with generalized elements: by the definition of preimage, for every generalized element x of T, we have

  $$x \in t^{-1}(t) \Leftrightarrow tx \in t$$

  But $tx = tx$, so $tx \in t$ for every x. So $t^{-1}(t)$ is the greatest subobject in $\mathrm{Sub}(T)$. Therefore, $t^{-1}(t) \sim \mathrm{id}_T \in \mathrm{Sub}(T)$ and $t = \chi_{\mathrm{id}_T}$.

Comparing these three proofs is very interesting. Personally, I like the second method the most: the Yoneda lemma gives us a strong, flexible and intuitive way to think about an object defined as the representing object of some functor. This covers examples like limits, exponential objects, the subobject classifier, and more. But in principle it can be wordy: one needs to justify that every step in the computation is natural and an isomorphism, write down how elements are sent, and check the properties.

7. Translating the greatest subobject [jc-000S]

- As a natural transformation $\{*\} \to \hom(-, \Omega)$, or as a universal element $1 \to \Omega$.
- Prove that $T \hookrightarrow \Omega$ and $1 \hookrightarrow \Omega$ are the same subobject, getting back to the Wikipedia definition.

In the previous section, we saw that the subobject classifier gives us the following data:

- a universal subobject $t : T \hookrightarrow \Omega$ (in fact, $t = \chi_{\mathrm{id}_T} : T \hookrightarrow \Omega$);
- for every subobject $u \in \mathrm{Sub}(A)$, a unique morphism $\chi_u : A \to \Omega$ such that $u \sim \chi_u^{-1}(t)$.

In this section, we want to see how the subobject classifier ***internalizes*** some special subobjects as morphisms in the underlying category. Externally, we have the following concepts:

- the equality of morphisms $f = g$,
- the membership $x \in u$ of a generalized element in a subobject,
- the logical "and", or the intersection subobject $u \cap v$.
  We claim that we can convert each of these concepts into a morphism $p : A \to \Omega$, and that we have a collection of specified morphisms $\mathrm{true}_A : A \to \Omega$, one for each A, such that $p = \mathrm{true}_A$ is equivalent to $f = g$, to $x \in u$, or to $q_1 = \mathrm{true}_A \wedge q_2 = \mathrm{true}_A$, respectively.

First, we focus on finding out what true is.

Recall the greatest subobject $\mathrm{id}_A : A \hookrightarrow A \in \mathrm{Sub}(A)$. We have seen that this construction is so uniform that the assignment

$$\begin{aligned} \underline{\{*\}} &\to \mathrm{Sub} \\ \{*\} &\to \mathrm{Sub}(A) \\ * &\mapsto \mathrm{id}_A \end{aligned}$$

forms a natural transformation. Now, using the concept of representable functors: the constant singleton functor $\underline{\{*\}}$ is represented by the terminal object 1, and the subobject functor Sub is represented by the subobject classifier Ω, so we have

$$\hom_{\mathcal{C}}(-, 1) \cong \underline{\{*\}} \to \mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega)$$

It sends $!_A \mapsto * \mapsto \mathrm{id}_A \mapsto \chi_{\mathrm{id}_A}$, and naturality says $\chi_{\mathrm{id}_B} = \chi_{\mathrm{id}_A} f$ for every morphism $f : B \to A$. By the Yoneda lemma, computing $\mathrm{id}_1 \mapsto * \mapsto \mathrm{id}_1 \mapsto \chi_{\mathrm{id}_1}$, we get a universal element $\chi_{\mathrm{id}_1} : 1 \to \Omega$ such that $\chi_{\mathrm{id}_A} = \chi_{\mathrm{id}_1} !_A$ for every object A.

Key Exercise 7.1. [jc-000T]
Notice that any morphism out of the terminal object is a monomorphism, so $\chi_{\mathrm{id}_1} : 1 \hookrightarrow \Omega \in \mathrm{Sub}(\Omega)$. Show that $x \in u \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$ for every subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A.

If you didn't shout "WOW" here, you should go back and do all three Key Exercises above.

Spoiler warning!!!

Both t and $\chi_{\mathrm{id}_1} \in \mathrm{Sub}(\Omega)$ have the property that for every $u \in \mathrm{Sub}(A)$ and every generalized element $x : \Gamma \to A$ of A, we have

$$x \in u \Leftrightarrow \chi_u x \in t$$

$$x \in u \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$$

Both act like "true"! Indeed, taking $u := t$ in the second equation, we get $x \in t \Leftrightarrow \chi_t x \in \chi_{\mathrm{id}_1}$; but $\chi_t = \mathrm{id}_\Omega$ by construction, so $t \sim \chi_{\mathrm{id}_1} \in \mathrm{Sub}(\Omega)$: they are indeed the same subobject!

Proof of equation 1: by the property of t, we have $u \sim \chi_u^{-1}(t)$. Expanding the definitions of $\sim$ and of preimages, we are done.
Proof of equation 2: first, we have $x \in u \Leftrightarrow x^{-1}(u) \sim \mathrm{id}_\Gamma$, since

$$x \in u \Leftrightarrow x\,\mathrm{id}_\Gamma \in u \Leftrightarrow \mathrm{id}_\Gamma \in x^{-1}(u) \Leftrightarrow \mathrm{id}_\Gamma \leq x^{-1}(u) \Leftrightarrow \mathrm{id}_\Gamma \sim x^{-1}(u)$$

where we used that $\mathrm{id}_\Gamma \leq x^{-1}(u)$ iff id_Γ factors through $x^{-1}(u)$, and that id_Γ is the greatest subobject.
Next, we show $x^{-1}(u) \sim \mathrm{id}_\Gamma \Leftrightarrow \chi_u x \in \chi_{\mathrm{id}_1}$. If $x^{-1}(u) \sim \mathrm{id}_\Gamma$, applying χ to both sides, we get

$$\begin{array}{ccc} \chi_{x^{-1}(u)} & = & \chi_{\mathrm{id}_\Gamma} \\ \| & & \| \\ \chi_u x & & \chi_{\mathrm{id}_1} !_\Gamma \end{array}$$

where the two vertical equalities use the naturality of $u \mapsto \chi_u$ and of $!_A \mapsto \chi_{\mathrm{id}_A}$. Therefore, $\chi_u x \in \chi_{\mathrm{id}_1}$.
Conversely, if $\chi_u x \in \chi_{\mathrm{id}_1}$, say $\chi_u x = \chi_{\mathrm{id}_1} y$ for some $y : \Gamma \to 1$, then again by naturality, $\chi_{x^{-1}(u)} = \chi_u x = \chi_{\mathrm{id}_1} y = \chi_{\mathrm{id}_\Gamma}$; therefore $x^{-1}(u) \sim \mathrm{id}_\Gamma$.

We should give the universal subobject a new name: $\mathrm{true} : 1 \hookrightarrow \Omega$, $\mathrm{true} := \chi_{\mathrm{id}_1}$; then $\mathrm{true} \sim t \in \mathrm{Sub}(\Omega)$. It can be understood as a global element of Ω, so it also corresponds to a collection of morphisms $\mathrm{true}_A : A \to \Omega$ (in fact, $\mathrm{true}_A = \chi_{\mathrm{id}_A}$ and $\mathrm{true}_1 = \mathrm{true}$) which is invariant under context substitution: $\mathrm{true}_\Gamma \sigma = \mathrm{true}_\Delta$ for every $\sigma : \Delta \to \Gamma$, and so

$$\mathrm{true}_A = \mathrm{true}\, !_A$$

for every object A.

Finally, rewriting the results above, we have

$$x \in u \Leftrightarrow \chi_u x = \mathrm{true}\, !_\Gamma$$

for every subobject $u \in \mathrm{Sub}(A)$ and generalized element x of A. Equivalently, the following

$$\begin{array}{ccc} U & \xrightarrow{!_U} & 1 \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \mathrm{true}} \\ A & \xrightarrow[\chi_u]{} & \Omega \end{array}$$

is a pullback square for every subobject $u \in \mathrm{Sub}(A)$.

8. Outro [jc-000U]

- Naturality: uniform constructions.
- Representable functors: internalizing natural transformations as morphisms.
- Exercise: prove that the uniform construction of the diagonal embedding is NOT a natural transformation (!!!), and prove that $x = y$ iff $\mathrm{eq}_A \langle x, y \rangle = \mathrm{true}_\Gamma$; this internalizes the equality of morphisms.
- Exercise: write the uniform construction of intersection as a natural transformation $\mathrm{Sub} \times \mathrm{Sub} \to \mathrm{Sub}$ and as a universal element $\wedge : \Omega \times \Omega \to \Omega$, and prove that $\wedge \langle p, q \rangle = \mathrm{true}_\Gamma$ iff $p = \mathrm{true}_\Gamma$ and $q = \mathrm{true}_\Gamma$; this internalizes the logical "and".

...

Exercise 8.1. [jc-000V]
Recall that the diagonal embedding $\Delta_A : A \hookrightarrow A \times A$, $\Delta_A := \langle \mathrm{id}_A, \mathrm{id}_A \rangle$, satisfies $x = y$ if and only if $\langle x, y \rangle \in \Delta_A$ for all generalized elements $x, y : \Gamma \to A$. Define $\mathrm{eq}_A : A \times A \to \Omega$ by $\mathrm{eq}_A := \chi_{\Delta_A}$. Show that

$$x = y \Leftrightarrow \mathrm{eq}_A\langle x, y\rangle = \mathrm{true}_\Gamma$$

We can understand $\mathrm{eq}_A\langle x, y\rangle : \Gamma \to \Omega$ as the internal proposition saying whether x and y are equal. Show that

$$\begin{aligned} \underline{\{*\}} &\to \mathrm{Sub}(- \times -) \\ \{*\} &\to \mathrm{Sub}(A \times A) \\ * &\mapsto \Delta_A \end{aligned}$$

is NOT a natural transformation (why!?).

Exercise 8.2. [jc-000W]
Recall that we have the intersection $u \cap v \in \mathrm{Sub}(A)$ of any subobjects $u, v \in \mathrm{Sub}(A)$, satisfying $x \in u \cap v \Leftrightarrow x \in u \wedge x \in v$. Show that

$$\begin{aligned} \mathrm{Sub} \times \mathrm{Sub} &\to \mathrm{Sub} \\ \mathrm{Sub}(A) \times \mathrm{Sub}(A) &\to \mathrm{Sub}(A) \\ (u, v) &\mapsto u \cap v \end{aligned}$$

is a natural transformation. Using $\mathrm{Sub} \cong \hom_{\mathcal{C}}(-, \Omega)$ and the Yoneda lemma, show that it induces a morphism $\mathrm{and} : \Omega \times \Omega \to \Omega$ satisfying

$$\mathrm{and}\ \langle p, q\rangle = \mathrm{true}_\Gamma \Leftrightarrow (p = \mathrm{true}_\Gamma \wedge q = \mathrm{true}_\Gamma)$$

for all morphisms $p, q : \Gamma \to \Omega$. We call a morphism $p : \Gamma \to \Omega$ an internal predicate on Γ, and we can understand $\mathrm{and} : \Omega \times \Omega \to \Omega$ as the internal connective "and".

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